Effective Knowledge Distillation on Reasoning Chain for Aspect-Based Sentiment and Emotion Analysis of Large Unannotated Data

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Abstract

Identifying the target of emotional words or phrases in crisis situations, especially health-related ones, is important for understanding public concerns. This paper applies aspect-level sentiment and emotion classification, a technique for identifying both specific topics and the sentiments and emotions expressed toward them. To handle the large volume of unannotated data, this study proposes a teacher-student knowledge distillation framework, where a teacher LLM generates reasoning explanations that guide a student model's training. The student (smaller) model learns five tasks from the framework: aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification. As such, the student model can be used on unannotated data for a complete ABSA and emotion detection pipeline, making it suitable for practical applications where communication practitioners often lack access to large-scale computational resources and annotated datasets. The code will be open sourced for future research.[1]

1 Introduction

During major health crises, such as the COVID-19 pandemic, public discourse becomes emotionally complex. Understanding this discourse requires more than identifying whether a statement is positive or negative; we must identify what specific topic (or "aspect") the emotion targets. For example, "fear" directed at the virus implies a willingness to cooperate with health measures, while "fear" directed at a vaccine may indicate a loss of trust in safety measures. This aspect-level analysis is crucial because aggregate sentiment measures obscure distinctions in public response. The same emotional state can signal opposite intentions depending on its target. While Aspect-Based Sentiment Analysis (ABSA), a set of techniques for identifying topics and their associated sentiments, has been extensively researched for e-commerce and product reviews, standard approaches focus on coarse sentiment polarities (positive/negative/neutral) rather than the fine-grained emotions (e.g. fear, anger, trust, hope) that drive behavior in crisis contexts. In this paper, we extend the task by jointly modeling aspect extraction, aspect based sentiment analysis, and aspect level emotion detection on unannotated COVID-19 tweets.

A significant barrier when doing analysis at aspect level is the lack of large COVID-19 datasets manually annotated for aspect based sentiment or emotion classification. While there are several COVID-19 datasets, they typically focus on named entities and coarse sentiment rather than specific emotion categories.

To avoid manual annotation, we propose a teacher–student knowledge distillation framework in which a large teacher model automatically annotates existing COVID-19 data and provides supervision for a smaller student model. Being orders of magnitude smaller, the student model can be used by researchers without access to high-performance GPU clusters.

Our modeling approach follows the Syn-Chain framework (Fan et al., 2025), which has demonstrated that, for the task of ABSA, incorporating explicit reasoning traces and auxiliary sub-tasks such as syntactic parsing and opinion extraction improves knowledge distillation performance compared to training only on labels. However, their approach is limited by the assumption of aspect-level annotation, hindering its applicability to unannotated data. We implement a five-stage reasoning pipeline. The teacher model generates labels and step-by-step reasoning traces for aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification. The

[1] https://github.com/onatakca/synchain-absa-emotion

outputs of the teacher model become supervision signals for a smaller, more efficient student model to learn from. Learning on both labels and reasoning traces allows the student model to learn the logic behind the classification rather than just the final label.

1.1 Research Questions and Contributions

Our work addresses the following questions:

- **RQ1**: Can synthetic annotations from LLM enable effective ABSA training on domain specific data, absent of aspect-level annotations done by humans?
- **RQ2**: Do explicit reasoning traces and auxiliary tasks in the knowledge distillation process improve the student model's performance on downstream tasks (aspect extraction, sentiment classification, emotion classification) compared to distillation using coarse labels?
- **RQ3**: How do aspect level emotions and sentiments differ in COVID-19 discourse, and what insights can this provide?

Our main contributions are: (1) extending the Syn-Chain reasoning pipeline with explicit aspect extraction and emotion classification steps; (2) creating a annotated COVID-19 ABSA dataset with emotion labels; and (3) demonstrating that the resulting model can serve both for downstream analysis, which can be used for (crisis) communication practitioners, and as an automatic annotator for new data.

2 Related Work

Our work draws on several research directions that together support aspect-level emotion analysis in unannotated data regimes with limited computational resources.

Weak Supervision and Automatic Labeling The challenge in interpreting tweets during some particular event, like COVID-19 is the scarcity of manually annotated data. To address this, researchers have turned to **weak supervision**, a technique where noisy or automatically generated labels are used for training. Weak supervision, via automatically generated pseudo-labels has show to be viable for aspect-based sentiment analysis. Notable performance gains are achievable even without manual annotations when syntactic structures are incorporated into the model (Negi et al., 2024). They showed that even without manual ground truth, fine-tuning a classifier on this synthetic data yields notable performance gains, provided that syntactic structures are incorporated.

Syntactic Dependency in ABSA Identifying the precise target of a sentiment requires understanding the grammatical relationships between words, which is the syntactic dependencies that link opinions to their targets.

Previous research showed that modeling syntax is essential for accurately disentangling complex sentiments in crisis discourse in both English and Chinese COVID-19 datasets Hou et al. (2025). Thus, our modeling framework includes syntactic parsing.

Emotion Detection in Crisis Discourse This study aims to capturing granular emotional categories (e.g., fear, anxiety, anger) because clearly identified emotional targets is crucial for understanding public health compliance and response (Abadi et al., 2021).

The identification of emotional targets is feasible. Previous research has shown that downstream models such as BERT can be trained using initial emotion annotations obtained from lexicon-based tools (e.g., Text2Emotion) Salsabila et al. (2023). While automated annotation establishes a baseline for aspect emotion classification, reliance on fixed lexicons limits a model's ability to capture context-dependent nuances such as sarcasm or implicit emotions. Our approach addresses this limitation by using contextualized reasoning from large language models.

Reasoning-Based Knowledge Distillation Recent advancements have moved beyond simple label prediction toward explicit reasoning, enabling models to learn not only what to predict but also why. This capability allows models to function as zero-shot downstream annotators, removing the need for manual aspect-level annotations. Using a teacher–student knowledge distillation approach, reasoning traces, step-by-step textual explanations of how decisions are reached, can be transferred from a large model to a smaller one Fan et al. (2025). Such an approach significantly outperforms standard supervised fine-tuning of BERT-based models on labels alone, demonstrating that reasoning chains provide valuable inductive bias for ABSA tasks (Fan et al., 2025). Our work builds directly on this framework, extending the reasoning chain to include aspect extraction and explicit emotion extraction, which are missing in prior models.

Instruction Tuning
Finally, the method of prompting the model (how we phrase the task description) influences performance. Cabello and Akujuobi (2024) introduced PFInstruct, showing that pre-pending NLP-specific task prefixes to instructions improves results across ABSA sub-tasks. This suggests that how the task is defined for the model is as critical as the model architecture itself, a principle we apply in our prompting design.

3 Proposed Method

3.1 Method Overview

To address the challenge of identifying emotional targets without large-scale manual annotation, our methodology proposes a Teacher-Student Knowledge Distillation framework. Our framework is shown in Figure 1, and the task logic is detailed in Figure 2.

The core idea is to use a large, capable language model's reasoning abilities to create high-quality training data for a smaller model. This approach leverages the reasoning capabilities of a massive Large Language Model (the Teacher) to synthesize training data that includes explicit reasoning traces explaining the logic behind each label. These traces are then used to fine-tune a smaller, more efficient Student model. This allows the Student to internalize the complex reasoning required for emotion detection, the task of learning *why* a sentiment is directed at a specific target, while maintaining the computational efficiency required for large-scale analysis.

3.2 Data Selection

We selected datasets COVID-19 tweet datasets that provide sentence level emotion labels for further pseudo annotation. Multiple COVID-19 Twitter datasets exist for sentiment analysis, including TweetsCOV19 (**?**), COVIDSenti (Naseem et al., 2021), COVID19 NLP (Miglani, 2020), TB-COV (**?**), and METS-CoV (Wang et al., 2022). Among them, METS-CoV (Wang et al., 2022) contains approximately 10,000 COVID-19 tweets annotated with medical entities and targeted sentiment, and was primarily designed for NER and entity level sentiment classification. COVIDSenti provides approximately 90,000 tweets labeled with coarse sentiment categories (positive, negative, neutral) (Naseem et al., 2021). However, many popular datasets, including METS-CoV (Wang et al., 2022), only release tweet IDs rather than the full text, which requires access to the Twitter/X API to retrieve the content. Since access has become prohibitively expensive, we focus on datasets that provide the complete tweet text: **COVIDSenti** (Naseem et al., 2021), and **COVID19 NLP** (Miglani, 2020) datasets.

COVIDSenti (Naseem et al., 2021) provides approximately 90,000 tweets labeled with coarse sentiment categories (positive, negative, neutral) for sentiment analysis, without aspect-level annotations. (Miglani, 2020) contains 48,000 tweets collected from Twitter and manually annotated at the sentence level with five sentiment labels (extremely positive, positive, neutral, negative, extremely negative).

Since these datasets lack specific emotion labels, we perform pseudo annotation using our Teacher model. We feed the raw sentences into the Teacher model to generate a dataset containing five reasoning traces (R_{1-5}) for every sentence. This synthetic dataset becomes the ground truth for training the Student.

News Filtering. A challenge in using these datasets for aspect-based sentiment analysis is the number of "news headline" style tweets, such as factual statements, news links, and quoted headlines that lack personal opinion or emotion, and therefore introduce irrelevant noise during training. Such tweets are unsuitable for ABSA, as they do not express sentiment toward specific aspects. Tweets identified as news were removed during preprocessing. News tweets were detected using patterns such as URLs, common headline-style prefixes, source attributions, and truncated text indicative of headline snippets.

Text Preprocessing. The remaining tweets underwent additional preprocessing, including emoji removal, Unicode normalization using the `ftfy` library, and URL stripping. We then performed syntactic parsing using spaCy's `en_core_web_sm` model, converting each tweet to CoNLL-U format, which encodes token-level information such as lemma, part-of-speech tags, and dependency relations.

Subdividing Dataset for Annotation. To enable efficient parallel annotation with the teacher model, we divided the datasets into balanced chunks, stratified by sentiment class to maintain representative distributions. This chunking strategy also enabled

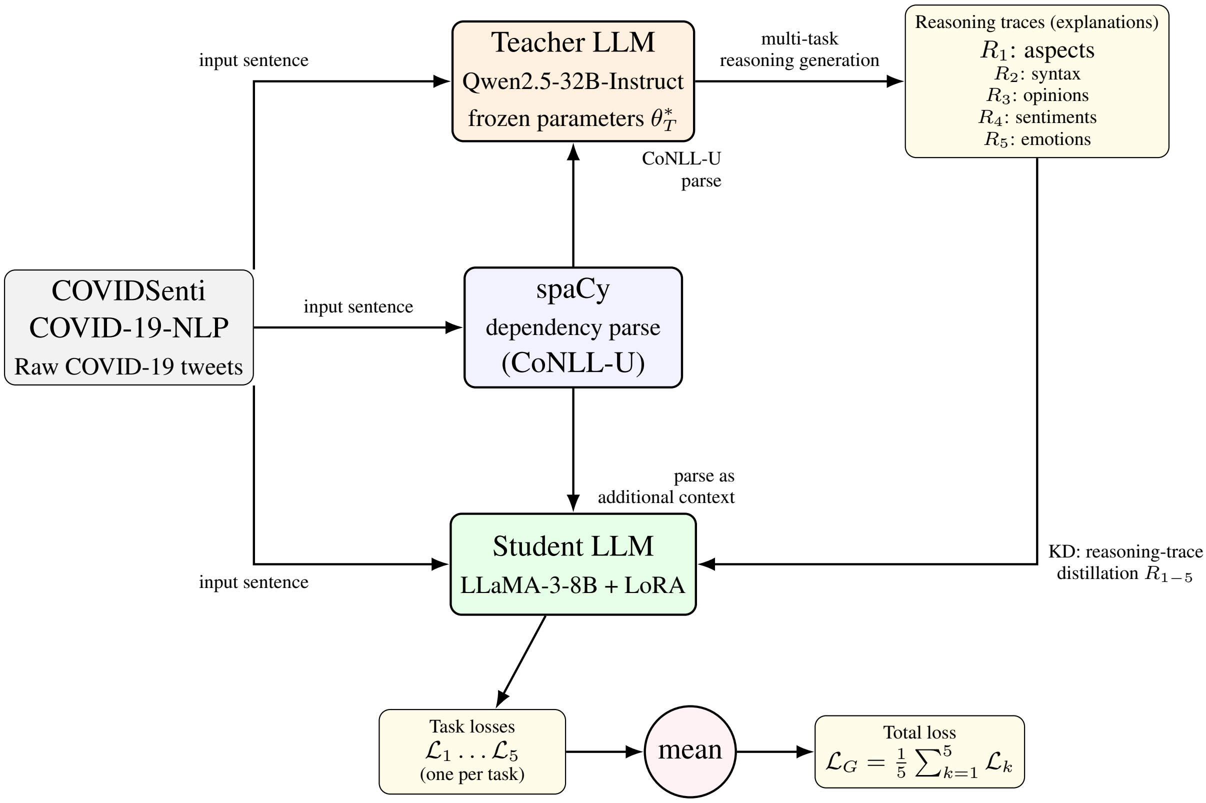

Figure 1: System overview of the proposed teacher–student framework. Each input sentence is parsed by spaCy into CoNLL-U format and, together with the raw sentence, is fed to a frozen teacher LLM with parameters θ_T^*. The teacher produces multi-task reasoning traces R_1–R_5 (explanations for syntax, aspects, opinions, emotion, and sentiment), which serve as knowledge-distillation (KD) targets for the student LLM (LLaMA-3-8B with LoRA). The student is trained with task-specific generative losses $\mathcal{L}_1 \ldots \mathcal{L}_5$, averaged into the total loss $\mathcal{L}_G$.

checkpointing and resumable annotation under limited compute resources.

3.3 Emotion Taxonomy

For aspect-level emotion classification, we adopt a 14-category emotion taxonomy designed for crisis discourse. The SenWave dataset (Yang et al., 2025) suggests the following seven categories: *optimistic*, *thankful*, *empathetic*, *pessimistic*, *anxious*, *sad*, and *annoyed*. We extend this set with seven additional categories derived from fine-grained emotion taxonomies used in prior work (Demszky et al., 2020): *hopeful*, *proud*, *trustful*, *satisfied*, *scared*, *angry*, and *neutral*. We believe this taxonomy captures the details in emotional responses characteristic of pandemic discourse, distinguishing between related but distinct states such as *anxious* versus *scared*, or *optimistic* versus *hopeful*. The inclusion of no_emotion allows the model to handle factual statements about aspects where no affective content is present.

Unlike sentence-level emotion detection, our task assigns exactly one emotion label per aspect, enabling fine-grained analysis of how different targets within the same tweet evoke distinct emotional responses. For instance, a tweet may express *thankful* toward healthcare workers while simultaneously expressing *anxious* toward vaccine side effects.

3.4 The Reasoning Pipeline

Our pipeline consists of five consecutive tasks. In this section, we describe each step and explain how the reasoning traces guide the student model's learning.

Step 1: Aspect Extraction. Building on the syntactic structure, we identify what entities or topics are being discussed. For each sentence, the model identifies aspect: specific entities, topics, or targets of opinion. For example, in "The vaccine rollout has been slow but the side effects are minimal," the aspects are "vaccine rollout" and "side effects." The teacher model generates a reasoning trace R_1 explaining why each span constitutes an aspect (e.g., "vaccine rollout" is identified because it represents a concrete policy action that opinions are expressed about). This task enables our student

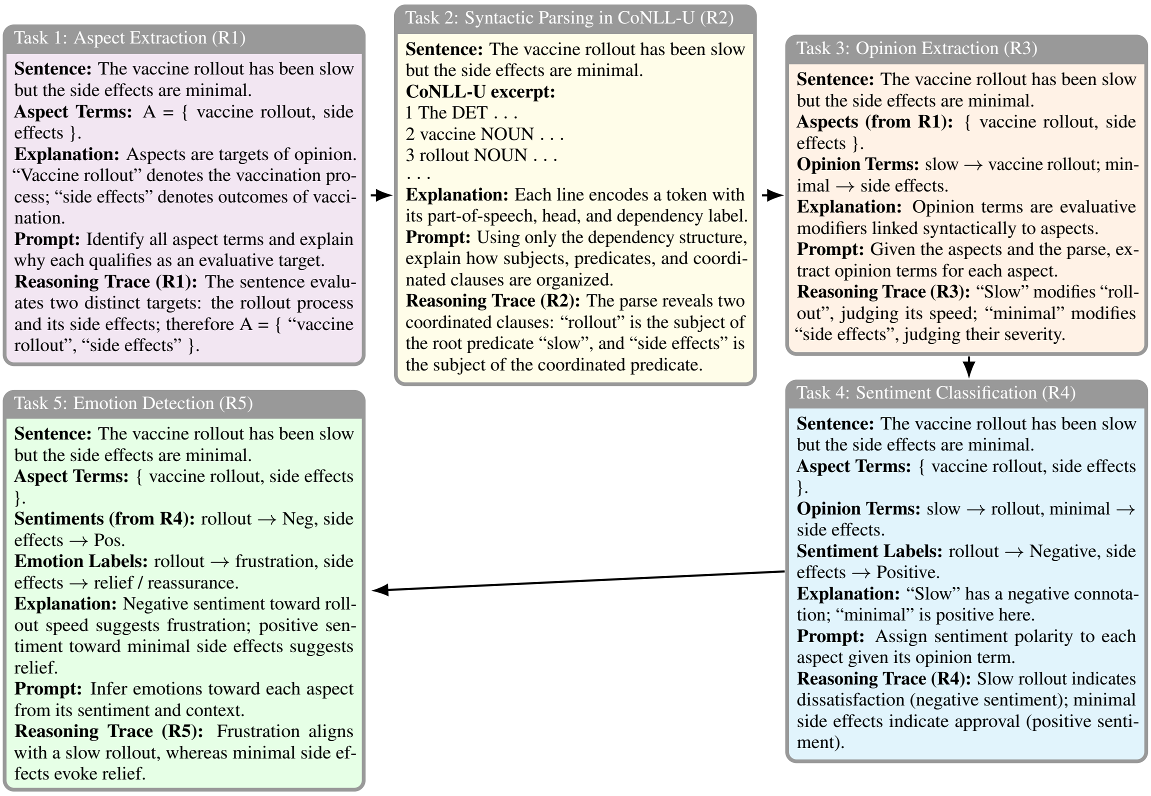

Figure 2: Reasoning example for aspect-based sentiment analysis and emotion detection on a single sentence via our teaching framework.

model to be zero-shot aspect annotator.

Step 2: Syntactic Parsing. Based on the grammatical structure of each sentence obtained in the first step, we use spaCy to obtain dependency parses for each sentence, converting them to CoNLL-U format, a tabular representation that explicitly encodes each word's part-of-speech, head word, and dependency relation. This structured format is easier for LLMs to process than raw dependency trees. The teacher model generates a reasoning trace R_2 for syntactic parsing, learning to predict this structure from raw text. This serves two purposes: it provides syntactic features that inform the other tasks, and it acts as an auxiliary task that improves the student model's linguistic understanding.

Step 3: Opinion Extraction. For each identified aspect, the teacher model extracts the opinion terms expressing sentiment toward that aspect. In our example, "slow" expresses opinion toward "vaccine rollout" and "minimal" toward "side effects." The reasoning trace R_3 explains how each opinion term connects to its aspect through dependency relationships in the sentence. This step would improve performance gain significantly Fan et al. (2025).

Step 4: Sentiment Classification. For each aspect-opinion pair, the model determines sentiment polarity (positive, negative, neutral). The reasoning trace R_4 explains why a particular sentiment label is appropriate given the opinion terms and broader context.

Step 5: Emotion Classification. For each aspect, the model identifies emotions expressed toward it (e.g., fear, anger, joy, trust, sadness). These categories are based on closed-set emotion taxonomies, which are relevant for crisis discourse. The reasoning trace R_5 explains the emotional content, which may be more subtle than sentiment polarity.

Five reasoning traces were obtained from the teacher model Qwen2.5-32B-Instruct (Qwen Team, 2024): R_1 for syntactic parsing, R_2 for aspect extraction, R_3 for opinion extraction, R_4 for sentiment classification, and R_5 for emotion extraction. The Qwen2.5-Instruct model family is reported to achieve state-of-the-art performance among open-weight models. We used the 32B variant due to hardware limitations; annotating our corpus on four NVIDIA L40 GPUs (48 GB each) required approximately 7 days.

The student networks are Llama-3-8B (Meta AI, 2024) and Qwen2.5-7B-Instruct (Qwen Team, 2024), selected based on their strong distillation performance reported by Fan et al. (2025). Both students are fine-tuned using a LoRA adapter (Schulman and Lab, 2025) via knowledge distillation from the teacher model's reasoning traces. We apply explicit teacher forcing between reasoning steps, i.e., step-level teacher annotations are provided as input rather than using the student's predictions from previous step. We adopt LoRA due to its efficiency and the computational cost of full fine-tuning; prior work has shown that LoRA can serve as an effective substitute for full fine-tuning while enabling efficient parameter updates (Schulman and Lab, 2025).

Loss definition The unified training objective across all five tasks is formulated as a generative loss:

$$\mathcal{L}_G = -\frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T}\log P(g_{i,t} \mid \hat{g}_{i,<t}, C_i), \quad (1)$$

where N denotes the number of training samples, T is the length of the output sequence for each sample, $g_{i,t}$ is the ground-truth token at position t for sample i, $\hat{g}_{i,<t}$ is the previously generated token sequence up to position $t-1$, and $P(g_{i,t} \mid \hat{g}_{i,<t}, C_i)$ is the conditional probability of generating $g_{i,t}$ given the context C_i (which encodes the input sentence, aspect, and task-specific prompt) and the prefix $\hat{g}_{i,<t}$ Fan et al. (2025).

Golden Set Evaluation The final fine-tuned model is used for zero-shot evaluation on the Chinese COVID-related data provided by Minsi Li, which is automatically translated into English using Yi-34B-Chat (01.AI, 2024). Yi-34B-Chat is a bilingual (Chinese - English) large language model trained by 01.AI on a large-scale multilingual corpus, and is therefore suitable for high-quality translation in this setting. A subset of this data will be manually annotated to serve as a gold standard for further evaluation. The final deliverable is a open sourced fine-tuned model that supports end2end aspect based sentiment and emotion analysis on English COVID-related data.

Our main methodological contribution is the extension of the syntactic–opinion–sentiment reasoning pipeline with **explicit aspect extraction** and **explicit emotion classification** steps. This design ensures that the resulting model can be used both for downstream ABSA and as an automatic annotator to generate aspect, opinion, emotion, and sentiment labels on previously unannotated data.

3.5 Dataset Statistics

We report statistics computed from the Qwen2.5-32B teacher annotations, aggregated over the COVID19NLP (Miglani, 2020) and COVID-Senti (Naseem et al., 2021) datasets. The resulting corpus contains 21,937 sentences. In total, the annotations include 50,615 aspect instances, corresponding to an average of 2.307 aspects per sentence (Table 1).

The aspect-level label distributions in Table 2 sum exactly to the total number of aspects (50,615). During aggregation, we detected 13 orphan labels (entries with missing aspect IDs in the aspect map) for both sentiment and emotion, from parsing errors in the teacher model's JSON output. These orphan labels represent 0.026% of the total aspects (13/50,615) and were excluded from all analyses and from the training data to maintain data integrity. The negligible proportion suggests these are isolated formatting errors rather than systematic annotation issues.

Table 1: Corpus summary (Qwen2.5-32B annotations).

Metric	Value
Total sentences	21,937
Total aspects	50,615
Average aspects per sentence	2.307

Table 2: Aspect-level sentiment distribution (N = 50,615).

Sentiment	Count	Percent
Neutral	26,249	51.87%
Negative	18,288	36.13%
Positive	6,078	12.01%

4 Experiments

We compare the performance of both student models on three tasks: aspect detection, sentiment classification, and emotion classification. Sentiment classification is a three-class problem (positive/negative/neutral), while emotion classification is a 15-class problem.

We validate the benefit of training on all five pipeline stages by removing components of the pipeline and evaluating the resulting model performance. This allows us to assess whether improvements in syntactic parsing or opinion extraction lead to improved downstream performance on the aforementioned classification tasks.

Table 3: Aspect-level emotion distribution (N = 50,615). No emotion was a label assigned when model failed to return a emotion label.

Emotion	Count	Percent
Neutral	21,608	42.70%
Annoyed	10,488	20.73%
Anxious	8,877	17.54%
Scared	1,552	3.07%
Hopeful	1,640	3.24%
Satisfied	1,017	2.01%
Empathetic	1,436	2.84%
Angry	1,014	2.00%
Optimistic	818	1.62%
Trustful	550	1.09%
Pessimistic	684	1.35%
Sad	449	0.89%
Thankful	262	0.52%
Proud	150	0.30%
No_emotion	70	0.14%

To prevent error propagation throughout the chain, at each step we train the student using teacher forcing; we are using teacher ground-truth outputs rather than student predictions for subsequent tasks. This enables meaningful evaluation of sentiment and emotion classification performance regardless of the student model's performance on aspect extraction, which is the most challenging component of the pipeline.

We additionally compare model performance before and after fine-tuning to validate the efficacy of our approach, using the teacher model outputs as reference and thereby measuring the alignment between teacher and student predictions.

4.1 Example Annotations

To illustrate the complete reasoning pipeline, we present three representative examples that demonstrate our model's capability to handle diverse COVID-19 discourse. These examples showcase the five-task reasoning chain (R1-R5) and highlight how the model disambiguates sentiment and emotion at the aspect level.

Figure 3 presents two contrasting cases that demonstrate the importance of aspect-level analysis. The first example illustrates the model's ability to correctly identify metaphorical usage of "Coronavirus" (referring to Corona beer rather than the virus), resulting in neutral sentiment and satisfied emotion despite surface-level ambiguity. The syntactic parsing (R2) and opinion extraction (R3) steps are crucial here, as they reveal the humorous context through dependency relationships. In contrast, the second example shows genuine health concern, where incomplete grammar and informal language ("wish coronavirus not that serious") are correctly interpreted as expressing negative sentiment and anxiety toward the pandemic situation. These examples demonstrate how the reasoning traces enable the model to navigate linguistic complexity, informal language, and contextual nuance that would challenge traditional sentiment analysis approaches.

Figure 4 demonstrates the model's multi-aspect capability on a more complex tweet containing two distinct evaluative targets. The model successfully identifies both "media censorship" and "coronavirus outbreak" as separate aspects, performing independent reasoning chains for each. While both aspects receive negative sentiment classifications (R4), the emotion labels (R5) differ: anger toward censorship policies versus fear regarding outbreak severity. This distinction is semantically meaningful since anger reflects frustration with human decisions (censorship), while fear responds to the health threat itself. The ability to extract and independently analyze multiple aspects within a single tweet is essential for understanding nuanced public discourse during crisis situations, where opinions often target multiple entities simultaneously with varying emotional valences.

These examples collectively demonstrate our model's core capabilities: handling informal social media language, disambiguating metaphorical and literal usage, correctly attributing sentiments to specific aspects, and identifying fine-grained emotions beyond simple polarity classification.

4.2 Training details

We use LoRA fine-tuning, a parameter-efficient fine-tuning method, where only a small subset of weights (adapters) is trained (Schulman and Lab, 2025). For fair comparison between the two student models, Meta-Llama-3-8B-Instruct and Qwen2.5-7B-Instruct, we use the same fine-tuning recipe. Specifically, we use a learning rate of 2×10^{-5}, mixed-precision training, batch size of 8, and gradient accumulation of 8, resulting in an effective batch size of 64. We use an adapter rank of 16 targeting projection layers. For Qwen2.5-7B-Instruct we train 40M parameters ($\sim$5%), while for Meta-Llama-3-8B-Instruct we train 42M parameters (also $\sim$5% of total parameters).

Example 1: Metaphorical Usage

INPUT TWEET:
"Hell, I have a 12 pack of Coronavirus in my fridge. It only makes you sick if you drink the whole thing."

REASONING_CHAIN_OUTPUT:
[R1] ASPECT:
"Coronavirus" (Metaphor)
→ Trace: "Refers to product/beer, not virus."

[R2] SYNTAX:
"Coronavirus" detected as proper noun modifying "pack" (compound noun phrase).

[R3] OPINION:
Trace: "Light-hearted joke... dismissive attitude."

[R4] SENTIMENT:
NEUTRAL (Context is humor, not health opinion).

[R5] EMOTION:
SATISFIED (Trace: "Joking suggests humor/joy").

Example 2: Genuine Health Concern

INPUT TWEET:
"wish coronavirus not that serious and the effected ppl number stop increase."

REASONING_CHAIN_OUTPUT:
[R1] ASPECT:
"coronavirus" (detected as key aspect)

[R2] SYNTAX:
[abbreviated for space]

[R3] OPINION:
Trace: "The speaker expresses a desire for the coronavirus situation to be less severe..."

[R4] SENTIMENT:
NEGATIVE (The sentiment... is negative. Because the speaker expresses a wish for the situation to be less severe...)

[R5] EMOTION:
ANXIOUS ("The speaker... affected individuals to stop...")

Figure 3: Example reasoning traces for two COVID-19 tweets demonstrating the complete annotation pipeline. Left: sarcastic reference requiring contextual sentiment disambiguation. Right: anxious expression of genuine health concern with negative sentiment.

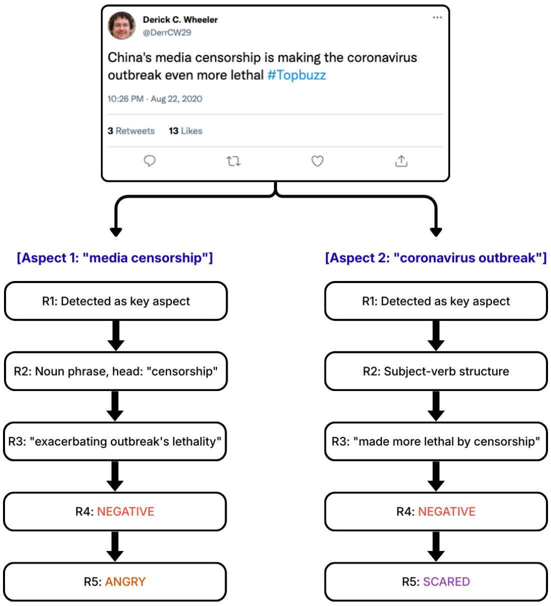

Figure 4: Reasoning trace across multiple aspects for a COVID-19 tweet. The model identifies two distinct aspects and performs complete annotation pipeline for each, resulting in shared negative sentiment but divergent emotions: anger toward media censorship versus fear regarding the outbreak itself.

4.3 Knowledge Distillation Results

Table 4 reports results for fine-tuning Llama-3-8B-Instruct and Qwen2.5-7B-Instruct. We evaluate four settings: (i) no fine-tuning, (ii) fine-tuning on downstream tasks only (3 tasks: aspect extraction, aspect-based sentiment classification, and emotion classification), (iii) fine-tuning on all five reasoning steps (adding syntactic parsing and opinion extraction), and (iv) ablations that omit one of the two additional reasoning tasks (syntax or opinion extraction) to test whether these traces improve student performance.

It is important to note that knowledge distillation in this setup is not fully model-agnostic: the token distribution and reasoning style are typically more similar within a model family than across families. Consequently, Qwen2.5-7B-Instruct may learn more effectively from the in-family teacher Qwen2.5-32B-Instruct than Llama-3-8B-Instruct.

Llama-3-8B-Instruct Results From Table 4, we draw the following conclusions for Llama-3-8B-Instruct: adding reasoning traces for syntactic parsing and opinion extraction does not provide meaningful benefit. The model achieves the best overall performance when fine-tuned only on the three downstream tasks (aspect extraction, sentiment classification, and emotion classification), reaching the highest aspect extraction F1 score of 64.8%, the highest sentiment classification accuracy of 92.9% across runs, and the second-highest emotion classification accuracy at 85.2%. Relative to the baseline, this corresponds to improvements of +23.5pp in aspect extraction F1, +1.8pp in sentiment classification accuracy, and +11.7pp in emotion classification accuracy.

Qwen2.5-7B-Instruct Results For Qwen2.5-7B-Instruct, fine-tuning with reasoning traces yields substantial gains over the out of the box baseline. The best overall configuration is obtained when fine-tuning on **all five tasks**, achieving the highest emotion classification accuracy (93.0%) and the strongest aspect extraction performance (F1 of

74.7%). Removing either syntactic parsing or opinion extraction slightly reduces overall performance, suggesting that the additional reasoning steps are beneficial for the Qwen student model. Relative to the baseline, best model improves aspect extraction F1 by 24.9%, emotion classification accuracy by 13.9%, while sentiment classification, even through high 91.0%, drops by 1.2% compared to the baseline.

Student Results Comparison Comparing the best-performing configurations of both students, Llama (3 task fine-tuning) versus Qwen (5 task fine-tuning), we observe that the Qwen student consistently performs better on the downstream objectives. In particular, Qwen achieves higher aspect extraction performance (F1 score of 71.2% vs. 64.8%) and significantly higher emotion classification accuracy (93.0% vs. 85.2%), indicating better alignment with the teacher outputs when distilling within the same model family.

4.4 Manual Golden Set Evaluation

We use tweets originally written in Chinese and automatically translated into English using Yi-34B-Chat. The dataset consists of 56,000 tweets. We apply both the teacher model (Qwen2.5-32B-Instruct) and the student model (Qwen2.5-7B-Instruct) and manually inspect 50 annotated samples from each model.

From Table 5, we observe that automatic translation is reliable (96% accurate), and that the student model achieves aspect recall consistent with results on our main dataset (70%).

For the teacher model, only a small number of aspects are missing or incorrect, and most predicted aspects align with old annotations, which supports the use of pseudo annotation as a substitute for costly human annotation.

The student model produces fewer aspects and had lower agreement with gold annotations than the teacher, but still demonstrates reasonable coverage. For both models, most sentiment and emotion predictions are correct.

Overall, manual evaluation validates our approach on an additional dataset and confirms that knowledge distillation transfers useful annotation behavior from teacher to student. It also reveals a limitation in the form of semantically overlapping aspects produced by both models. Future work will focus on improving aspect granularity and alignment with the target task.

5 Discussion

With respect to the research questions introduced in Section 1.1, we organize the following sections to show how our findings support each question.

5.1 Synthetic Teacher Annotations

We evaluate whether annotations produced by a high capacity teacher model can replace costly human annotations and serve as supervision for subsequent knowledge distillation to a smaller student model.

We observe that the teacher model consistently produces valid label sets, does not hallucinate, and follows the annotation prompts as specified. Because the data generation process remains reliable, the student model is able to learn from the pseudo labels and accompanying reasoning traces and successfully imitate the teacher model.

We also identify a limitation of the pseudo annotation process. The teacher model in some cases produces semantically overlapping lists of aspects. Since our study prioritizes recall over precision in order to preserve a rich training signal, and given the time cost of large scale annotation, we did not attempt to further refine or regenerate the annotations beyond initial prompt tuning. As a result, some sentences contain semantically overlapping or redundant aspects.

This limitation does not undermine the validity of our findings but points to a direction for future work, where improved prompting or post processing strategies could produce more precise aspect annotations.

5.2 Effectiveness of Knowledge Distillation

We evaluate the effectiveness of our framework by comparing against frozen baselines and multiple fine-tuning settings, including fine-tuning only on downstream tasks (aspect extraction, sentiment classification, and emotion classification) and fine-tuning with additional reasoning traces for syntactic parsing and opinion extraction.

Our results show that both student models, Meta-Llama-3-8B-Instruct and Qwen2.5-7B-Instruct, achieve substantial improvements across aspect extraction, sentiment classification, and emotion classification. For Llama-3-8B-Instruct, the best configuration score a +23.5pp improvement in aspect extraction F1, a +1.8pp improvement in sentiment classification accuracy, and a +11.7pp improvement in emotion classification accuracy over the

Table 4: Comparison of student model performance across training scenarios. We report aspect extraction precision/recall/F1, sentiment classification accuracy, and opinion extraction accuracy.

Model	Scenario	Aspect Extraction			Sentiment Cls.	Emotion Cls.
		Prec.	Rec.	F1	Acc.	Acc.
Meta-Llama-3-8B-Instruct	No fine-tuning	0.341	0.515	0.411	0.911	0.733
	Fine-tune (5 tasks)	0.649	0.622	0.635	0.927	0.829
	Fine-tune (4 tasks, w/o syntactic parsing)	0.629	**0.646**	0.638	0.926	**0.852**
	Fine-tune (4 tasks, w/o opinion extraction)	0.653	0.626	0.639	0.927	0.836
	Fine-tune (3 tasks, w/o opinions and syntax)	**0.654**	0.643	**0.648**	**0.929**	0.850
Qwen2.5-7B-Instruct	No fine-tuning	0.466	0.461	0.463	0.922	0.791
	Fine-tune (5 tasks)	0.681	**0.747**	**0.712**†	0.910	**0.930**†
	Fine-tune (4 tasks, w/o syntactic parsing)	0.665	0.723	0.693	0.920	0.920
	Fine-tune (4 tasks, w/o opinion extraction)	**0.696**	0.728	0.712	0.900	0.910
	Fine-tune (3 tasks, w/o opinions and syntax)	0.638	0.658	0.648	**0.929**†	0.852

Table 5: Manual evaluation of 50 samples. We report missing, wrong, and semantically overlapping predictions for teacher (total 185 aspects) and student (total 129 aspects) model across tasks.

Task	Model	Miss	Wrong	Overlap
Aspect Extraction	Teacher	4	1	10
	Student	13	17	26
Sentiment Cls.	Teacher	6	9	8
	Student	0	13	0
Emotion Cls.	Teacher	1	6	0
	Student	1	5	0

non fine-tuned baseline. Similarly, Qwen2.5-7B-Instruct achieves large gains in aspect extraction (+27.9pp f1) and emotion classification (+13.9pp accuracy), confirming the effectiveness of knowledge distillation with reasoning traces.

To assess the contribution of syntactic parsing and opinion extraction as part of our pipeline, we train models using all five tasks and conduct ablations that omit one or both of these auxiliary tasks. For Meta-Llama-3-8B-Instruct, adding syntactic parsing and opinion extraction does not lead to additional gains. We attribute this to cross-architecture distribution mismatch, where reasoning traces generated by Qwen2.5-32B-Instruct may not align well with the internal representations of Llama-3-8B-Instruct, or where limited training capacity is spent modeling the teacher reasoning style rather than improving downstream performance. Future ablation on longer training time would be needed to prove efficacy of this cross architecture setting.

In contrast, for the Qwen2.5-7B-Instruct student model, both syntactic parsing and opinion extraction provide measurable benefits. Fine-tuning on all five tasks results in the best overall performance, proving that additional reasoning steps are indeed meaningful when teacher and student models belong to the same model family.

Overall, these results support claim that the proposed reasoning pipeline is effective and that incorporating all five tasks can improve student performance. Exploring additional model architectures and teacher - student pairings remains as direction for future work to better understand cross-architecture knowledge transfer.

5.3 Emotion Patterns in COVID-19 Discourse

Based on the teacher annotations presented in Table 1 and Table 3, we observe that the majority of aspect-level sentiments are neutral (51.87%), and that 42.70% of aspect-level emotion labels are also neutral. At the sentence level, where annotations are available, we aimed to maintain similar label distributions.

It is possible for a sentence to exhibit an overall positive or negative valence while only the main aspect is assigned a corresponding sentiment and emotion label. Other, less central aspects that function as auxiliary nouns may be labeled as neutral because no explicit sentiment or emotional valence is expressed toward them. As a result, the distribution of neutral sentiment and emotion labels is skewed.

Another notable observation is the prevalence of negative sentiment labels, which occur approximately three times more frequently than positive labels, and negatively associated emotions, which appear about five times more often than positive ones. This pattern reflects the general negativity present in pandemic discourse. In crisis situations, such as COVID-19 pandemic, it is therefore particularly important to identify the specific targets of negative emotions, which highlights the value of our aspect-level analysis.

5.4 Limitations

Several limitations should be acknowledged. First, the approach relies entirely on teacher model annotations, meaning that any biases or errors in the teacher outputs may propagate to the training data. Although we conduct limited manual validation, the overall annotation quality across the full corpus remains to be verified.

Second, while the 14-category emotion taxonomy is comprehensive, it may not capture all nuances of emotional expression in crisis discourse.

Finally, the computational requirements for running the teacher model (Qwen2.5-32B) are substantial, which may limit scalability for very large datasets. However, the student models are considerably more efficient for inference, and quantized versions can run on low cost hardware.

5.5 Practical Implications

Despite these limitations, the approach has several practical applications. The fine-tuned student models are efficient enough for low cost deployment, enabling monitoring of public sentiment during health crises. This capability can help public health officials identify emerging concerns and respond accordingly.

The ability to generate domain-specific ABSA datasets without manual annotation significantly reduces both the time and cost of system development. Manual verification proves the reliability of the teacher pseudo annotations.

The explicit reasoning traces provide interpretability valuable for debugging and system refinement. This transparency makes it easier to identify and correct systematic errors compared.

6 Conclusion

We extend the Syn-Chain framework (Fan et al., 2025) to perform aspect-based sentiment analysis with emotion detection on unannotated COVID-19 tweets. Our primary contribution is the addition of explicit aspect extraction and a 14-category emotion taxonomy into the reasoning pipeline. Using Qwen2.5-32B as a teacher model, we automatically annotate 21,937 tweets with 50,615 aspect-level labels and fine-tune smaller student models (Meta-Llama-3-8B-Instruct and Qwen2.5-7B-Instruct) using LoRA.

Our main contributions are:

- We expand the knowledge distillation framework with aspect extraction and emotion classification, creating a five task reasoning pipeline (aspect extraction, syntactic parsing, opinion extraction, sentiment classification, and emotion classification) for aspect level sentiment and emotion analysis on unannotated data.
- We construct a COVID-19 ABSA dataset containing 50,615 annotated aspects across 21,937 tweets.
- We demonstrate that LoRA-based fine-tuning within our knowledge distillation framework significantly improves student model performance, with Qwen2.5-7B-Instruct achieving up to a 24.9% F1 improvement in aspect extraction and a 13.9% improvement in emotion classification.

Future Work During manual evaluation of teacher annotations, we observed that our teacher model produces semantically overlapping aspect predictions. To improve aspect granularity/precision, alternative prompting techniques, manual annotation of a small set of samples for in-context learning, and post-processing verification with a smaller model could be explored.

Second, future work should explore additional teacher–student architecture pairings and longer training schedules to better explore cross-architecture knowledge transfer. Our results indicate that within same compute budget student models benefit more when distilling from a teacher within the same model family. While syntactic parsing and opinion extraction benefit the Qwen2.5-7B-Instruct student model, they do not consistently improve performance for Llama-3-8B-Instruct.

Finally, expanding manual evaluation to larger and more diverse datasets, as well as extending the framework to additional domains and languages, would further validate the generality of the proposed approach.

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